Predicting vortex shedding behind a circular cylinder with physics-informed neural networks

Peeyush Agarwal
AE496 Undergraduate Project

September 22, 2026

Abstract

A data-free physics-informed neural network (PINN) applied to the unsteady wake of a circular cylinder is known to collapse onto the steady symmetric solution: the steady field satisfies the governing equations exactly, and the training loss has no notion of its physical instability. This project reproduces that collapse on purpose-generated, validated ground truth, and then determines which of three candidate mechanisms defeats it. On a lattice Boltzmann dataset at Re = 100 (St within 3.3% of published benchmarks, with a C_d overshoot of 4.8% traced to domain confinement), a plain PINN retains 10.6% of the reference wake amplitude. Adding Fourier features or a Reynolds decomposition does not help. *Causal time-marching alone restores shedding data-free*, at 89.2% retained amplitude, and an equal-budget control rules out extra optimiser steps as the cause. The result holds at every Reynolds number tested from 60 to 180: the control keeps 10–43% of the wake, the mechanism-equipped network 88–104%. The best data-free configuration, marched over five shedding periods, predicts the shedding frequency to within 0.1%, at 94.8% of the correct amplitude and 2.4° of phase, with a velocity-field error of 3.6%. A second contribution adds a pressure-free physics residual to a Fourier Neural Operator trained after the method of Renn et al.; the physics term lowers forecast error by 21–28% out to one shedding period, after which both variants have failed.

1 Introduction and objective

Above Re $\approx$ 47 the wake behind a circular cylinder becomes unstable and sheds a Kármán vortex street at a frequency characterised by the Strouhal number St $= fD/U$. Predicting that shedding with a neural network is not merely a question of accuracy, because of a structural difficulty:

1. **The steady solution remains a valid solution.** A symmetric, non-shedding field satisfies the Navier–Stokes equations and every boundary condition exactly. It is unstable physically, but a PINN loss measures residuals, not stability, so the steady field is an easily reachable minimum.
2. **Spectral bias.** Networks fit low-frequency content long before high-frequency content, which directly opposes representing an oscillatory wake.
3. **The initial condition is a measure-zero slice** of the space–time domain, and is therefore weakly weighted against the bulk PDE residual that fills it.

The documented consequence [1] is that a data-free PINN "behaves like a steady solver", while a data-driven one sheds only where data is present. Two points of scope. Their cylinder case is at Re = 200, solved as a *single* time window spanning forty shedding periods, whereas this project reproduces the phenomenon at Re = 100 — and across Re = 60–180 in Section 7

— from a baseline that solves one window of one period and collapses anyway, which is a considerably more favourable configuration than theirs. And they attribute the collapse to spectral bias, on the evidence of a Koopman analysis showing damped modes; the ablation below finds that attacking spectral bias directly does not restore shedding, while enforcing causality does, without touching the spectrum.

The objective is therefore to reproduce that failure on validated ground truth of our own, and then to establish *which* mechanism defeats it, with an ablation controlled well enough that the answer is attributable. The mechanism that works, causal time-marching, is not new in itself — it is the sequence-to-sequence formulation of Krishnapriyan et al. [4], demonstrated there on one-dimensional convection, reaction and reaction–diffusion problems, and a hard-window relative of the causal weighting of Wang et al. [5]. What is established here is that it suffices *alone*, *data-free*, on two-dimensional wake shedding, and that the gain is causality rather than compute. That matters because two independent studies conclude the opposite: both Chuang and Barba [1] and, more recently, Shah [3] — who ablates six techniques on a confined cylinder benchmark and finds that only supervised combinations escape the collapse — report that some form of data supervision is necessary to sustain shedding at all. Neither tests hard time-window marching.

A second thread connects the work to its reference paper. Renn et al. [2] forecast subcritical cylinder wakes with a Fourier Neural Operator (FNO) that is *purely data-driven* — there is no physics term anywhere in it. Whether adding physics to such an operator improves forecast stability is the natural question joining the two topics, and is addressed in Section 8.

Reynolds number. The ablation and the accuracy study are at $\mathrm{Re} = 100$: laminar and unambiguously two-dimensional. Renn et al.'s $\mathrm{Re} = 240$–3060 is genuinely three-dimensional, where a 2-D model would be solving the wrong physics. Section 7 repeats the central comparison across $\mathrm{Re} = 60$–180, the upper limit set by the three-dimensional transition at $\mathrm{Re}_c = 188.5 \pm 1.0$ [13].

Software. The work builds on `underPINN`, a JAX PINN framework, which supplies the network architectures, the geometry primitives and a steady Navier–Stokes residual. The unsteady operator (Section 3), the lattice Boltzmann solver, the ablation machinery and the operator-forecasting code are contributions of this project.

2 Ground truth by lattice Boltzmann

No experimental data was available, so the reference wake is generated. The solver (`src/cfd/lbm_cylinder.py`) is a D2Q9 two-relaxation-time (TRT) lattice Boltzmann method written in JAX.

Why these particular choices.

- **TRT collision.** A single-relaxation-time (BGK) scheme places the bounce-back wall at a viscosity-dependent position, so the cylinder's effective diameter drifts with Re. TRT relaxes the symmetric and antisymmetric parts separately, allowing the magic parameter $\Lambda = (\tau^+ - 1/2)(\tau^- - 1/2)$ to be pinned at $3/16$, which fixes the wall location.
- **Half-way bounce-back** for the no-slip cylinder, placing the wall midway between nodes.
- **Momentum-exchange** for the surface force, which is what makes C_d accurate enough to validate against published values.
- **Specular (free-slip) top and bottom walls**, enforcing $v = 0$ exactly and matching the symmetry condition used on the PINN side.

Shedding is triggered by a brief transverse inlet perturbation; without it the symmetric base flow persists indefinitely — the same degeneracy that the PINN later falls into, appearing here in the reference solver.

Production dataset. Re = 100, $D = 24$ lattice units, Ma = 0.087, grid 720×1200, blockage $D/H = 2\%$. The saved record is 641 snapshots spanning $t^* = 91$–212 on a 241×121 grid at $\Delta x = 0.083D$ and $\Delta t^* = 0.19$, chosen near the reference paper's PIV resolution so that the later comparison is like-for-like.

Table 1: Validation of the ground truth at Re = 100. The benchmarks are the compilation of eight independent large-domain studies in Qu et al. [11], whose own values (used as the reference for the errors quoted) are $C_d = 1.317$, St = 0.1649 and $C_l' = 0.2224$ at $H = 120$.

quantity	computed	benchmark [11]	error
C_d (mean drag coefficient)	1.3798	1.318–1.336	+4.8%
St (Strouhal number)	0.1595	0.164–0.166	−3.3%
divergence error	1.35% of $\|\nabla \mathbf{u}\|$, against $\mathrm{Ma}^2 = 0.75\%$		

Three points about this table, because each of them was got wrong at some stage of this project.

First, C_d is *not* within a fraction of a percent of the literature, as an earlier draft claimed against an uncited benchmark of 1.37–1.38. The large-domain consensus at Re = 100 is $C_d \approx 1.32$, and the overshoot here is +4.8%. It is systematic rather than incidental: the same comparison across the whole sweep of Section 7 gives +2.7% to +7.9% at every Reynolds number. Section 7 attributes it to confinement, and the convergence study below is the direct evidence.

Second, the St quoted is measured from wake vorticity, not from the lift signal, for the reason given in Section 7.1. The lift-derived value for this dataset is 0.1611, which would read as a −2.3% error and flatter the solver by a point.

Third, the divergence error depends on the evaluation mask to within an order of magnitude, because centred stencils adjacent to the cylinder straddle the wall and measure a stencil artefact rather than compressibility. The figure quoted uses this project's own `interior_mask` with a two-cell halo and an r.m.s. norm; widening the halo to eight cells gives 0.53%, and a mean-absolute norm gives 0.71%. The claim worth making is that the residual divergence is of order Ma^2, not that it takes any particular value.

Convergence study. Two candidate error sources were separated (Figure 1). Varying resolution from $D = 16$ to $D = 40$ at fixed blockage moves C_d by only 2.4%: the grid is converged. Varying blockage at fixed resolution moves it far more, $C_d = 1.627$, 1.458 and 1.431 at 5.0%, 3.3% and 2.5%, extrapolating linearly through the two least confined cases to 1.348 at zero blockage — still +2.3% against the benchmark, since the production dataset's $H = 50$ is itself well inside the confined regime. *The apparent +24% error at 5% blockage is domain confinement, not a solver defect* — the headline validation result, and one the literature supports directly: Qu et al. [11] report that force and base-suction coefficients depend more strongly on domain size than on resolution while St does not, and that blockage effects become negligible only for $H > 100$.

A methodological note. An early estimate of St taken from an FFT peak gave a fraction-of-a-percent agreement that was pure luck: over a record of this length a single FFT bin is wider than the discrepancy being measured, so the peak cannot resolve the quantity the agreement was claimed for. The St values reported here are measured instead in the time domain, from

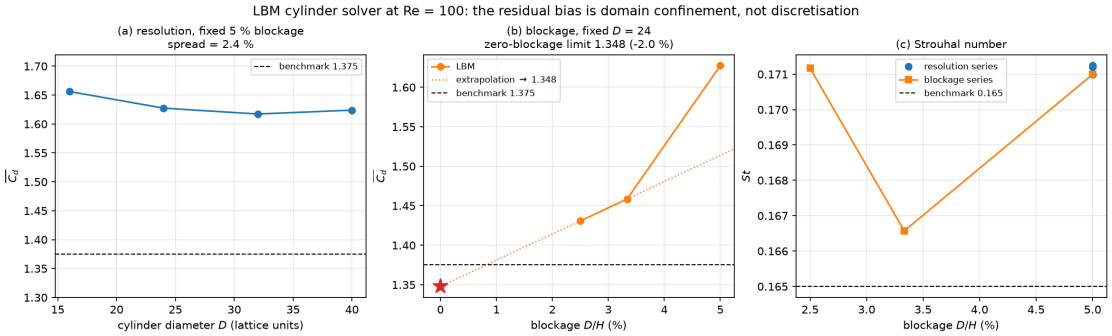


Figure 1: Solver validation: resolution series, blockage series, and the extrapolation to zero blockage.

a zero-crossing estimator that resolves the period to a fraction of a time step, cross-checked spectrally where the signal is clean enough (Section 7.1); the honest error is -3.3%.

3 An unsteady Navier–Stokes operator

`underPINN` ships only a steady incompressible residual, which cannot represent shedding at all. This project supplies the unsteady operator (`src/pinn/ns2d_unsteady.py`), conforming to the framework's `BasePDE` contract so it composes with the existing solvers, samplers and networks. Non-dimensionalised on D and U,

$$u_x + v_y = 0, \tag{1}$$

$$u_t + uu_x + vu_y + p_x - \tfrac{1}{\mathrm{Re}}(u_{xx} + u_{yy}) = 0, \tag{2}$$

$$v_t + uv_x + vv_y + p_y - \tfrac{1}{\mathrm{Re}}(v_{xx} + v_{yy}) = 0, \tag{3}$$

with the network mapping $(x, y, t) \mapsto (u, v, p)$.

Two derivative backends are implemented: a literal `jax.hessian` reference, and a forward-over-forward JVP path computing only the six derivative groups the residual needs, where a full Hessian would compute 27.

Table 2: Verification of the unsteady operator. All three checks pass at machine precision.

check	result
annihilates the Taylor–Green vortex (an exact NS solution)	1.6×10^{-16}
fast JVP backend versus literal Hessian	2.8×10^{-17}
steady limit versus `underPINN`'s own `NavierStokesPDE`	5.6×10^{-17}
speed-up of the fast backend	$2.5\times$

The third check is the important one: it shows the new operator agrees with the framework's existing physics, rather than being a parallel implementation that merely looks correct.

4 Reproducing the collapse

A plain multilayer perceptron $[3, 128^{\times 6}, 3]$ is trained on physics, boundary and initial conditions with *no data*, over a single time window of one shedding period, for 20 000 epochs.

Shedding retained is this project's own metric: the ratio of transverse-velocity r.m.s. at a fixed wake probe between the PINN and the LBM reference. Near 0% means collapse to steady flow; 100% means the wake amplitude matches.

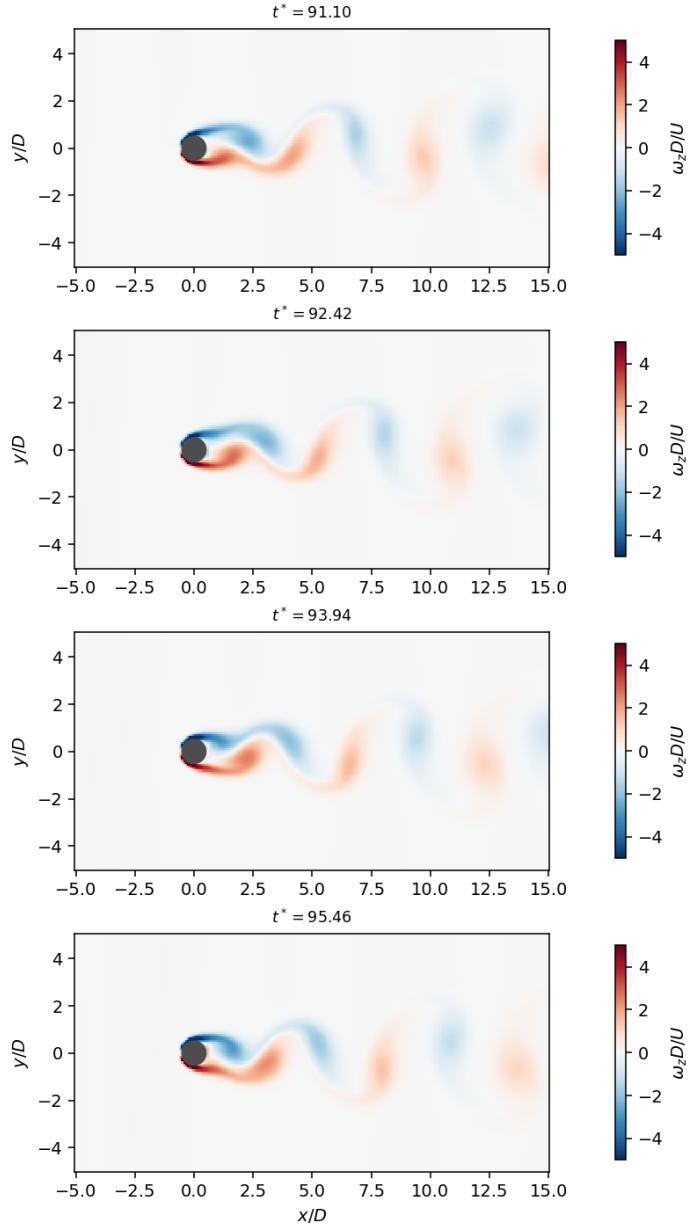


Figure 2: The reference Kármán street over one shedding period.

The wake keeps about a tenth of the reference amplitude — the PINN is effectively solving the steady problem. Quadrupling the budget to 80 000 epochs moves this only to 21.8%, so the collapse is not an artefact of under-training.

5 Defeating the collapse

Three mechanisms are implemented as switches on one trainer, so that the ablation is controlled — every configuration shares the sampler, the optimiser, the schedule and the evaluation.

Two implementation points matter. First, the Reynolds decomposition requires a *differentiable* mean field, because the momentum residual contains ∇u_0 and $\nabla^2 u_0$; bilinear interpolation of a grid gives piecewise-constant first derivatives and identically zero second derivatives. The mean is therefore fitted by a separate frozen network (relative $L_2 = 1.1 \times 10^{-3}$). Second, that mean is obtained by *regression on the measured mean*, not by a steady PINN solve, because the time mean of a shedding flow does not satisfy the steady equations — it carries the divergence

Table 3: The documented failure, reproduced. Two seeds.

quantity	seed 0	seed 1
shedding retained	9.8%	11.3%
relative L_2 on (u, v)	0.184	0.183

Table 4: The ablation matrix: what each configuration attacks.

configuration	mechanism	obstacle attacked
`baseline`	—	(control)
`fourier`	random Fourier features	spectral bias
`decompose`	Reynolds decomposition $u = u_0 + u'$	conditioning, dynamic range
`march`	causal time-marching	weak initial condition, causality
`all`	all three	—
`all+data`	plus 4 000 supervised points per window	—

of the Reynolds stresses.

5.1 Result

Table 5: Full-budget ablation at Re = 100: 20 000 epochs per window, two seeds, seed means. *Time-marching alone is sufficient, data-free.*

configuration	shedding retained	relative L_2
`baseline`	10.6%	0.183
`fourier`	9.7%	0.167
`decompose`	7.0%	0.110
`march`	**89.2%**	0.078
`all`	**90.2%**	0.047
`all+data`	100.0%	**0.015**

Neither spectral bias nor conditioning was the binding constraint. Both mechanisms reduce the field error — the decomposition notably, from 0.183 to 0.110 — while leaving the wake collapsed. What restores shedding is enforcing causality: solving a short window, then using its end state as the initial condition for the next, which strengthens the initial condition relative to the bulk residual and forbids information from flowing backwards in time.

5.2 The equal-budget control

A marched run of four windows takes four times the optimiser steps of a single-window run, so the comparison above is confounded by compute. Re-running the single-window configurations at 80 000 epochs — the same total — settles it:

This control is essential to the claim and should be quoted with it; without it, "time-marching wins" is indistinguishable from "time-marching trains longer".

5.3 Hyperparameters, and one that mattered only in combination

The Fourier bandwidth σ is worth several points *on top of* the mechanism that works, and nothing at all without it. The initial-condition weight, by contrast, changes nothing once

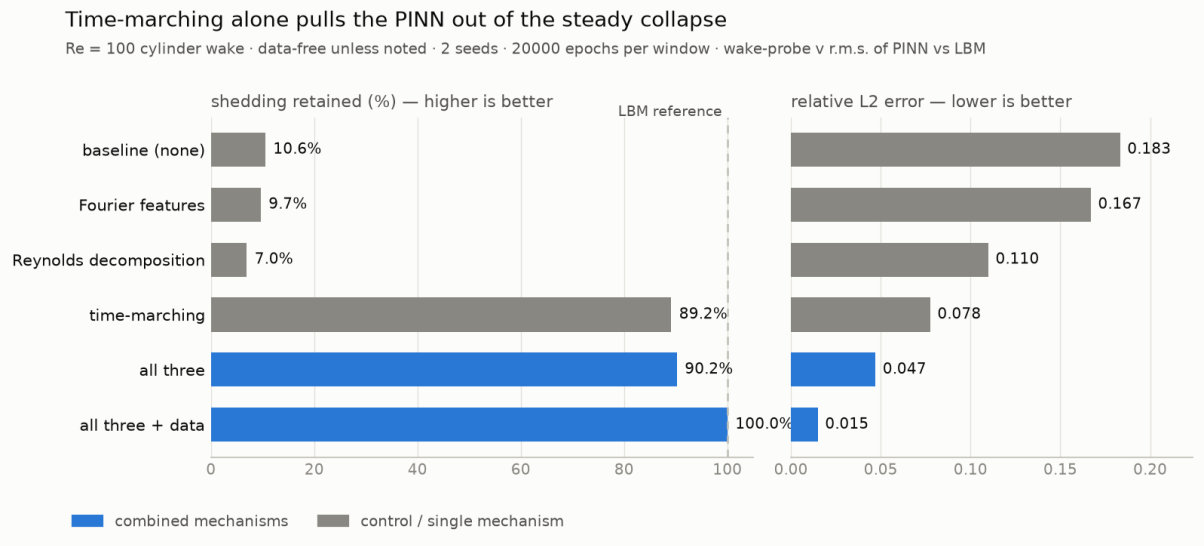


Figure 3: The ablation. Neither Fourier features nor the Reynolds decomposition escapes the collapse; causal time-marching does.

Table 6: Equal-budget control. Both improve; neither approaches 89%. *The gain is causality, not compute.*

configuration	20 000 ep, 1 window	80 000 ep, 1 window	$4 \times 20\,000$ ep, 4 windows
`baseline`	10.6%	21.8%	—
`decompose`	7.0%	23.7%	—
`march`	—	—	**89.2%**

marching supplies the causal chain — consistent with the interpretation that marching works by strengthening the initial condition, making further weighting redundant.

5.4 How much supervision is actually needed

Shedding saturates by 100 points per window; everything beyond that buys accuracy only. *The physics buys the dynamics; the data buys the last factor of three in error.*

6 How accurately does it shed?

Shedding retained is an amplitude ratio: it answers "is there a wake", which is the right question for the collapse and the wrong one for prediction accuracy. A network could hold the right amplitude at the wrong frequency, or drift out of phase within the window, and the ablation table would not notice. The best data-free configuration was therefore compared against the reference probe signal directly.

Each column is scored against the LBM signal over its *own* span, which is why no single reference value is quoted: the one-period window covers $\mathrm{St}_{\mathrm{LBM}} = 0.1578$ and the five-period window 0.1625, the difference being what ten zero crossings buy over two.

Extending the run from four windows to twenty — from one shedding period to five — *improves* every quantity that matters. The frequency error falls to +0.1% on ten zero crossings rather than two, the field error drops from 0.043 to 0.036, and the phase drifts by 0.005 of a period (about 2°) across the entire five-period span. The marching chain therefore does not accumulate error in any visible way over the horizon tested: the wake stays locked to the reference in both frequency and phase, and only the amplitude gives up a little (96.5% to 94.8%).

Table 7: Knobs, two seeds each.

change	applied to	shedding retained	relative L_2
$\sigma = 8$ (from 2)	`all`	90.2% → **97.6%**	0.047 → **0.042**
$\sigma = 8$	`fourier`	9.7% → 11.9%	0.167 → 0.148
$w_{ic} = 100$ (from 10)	`all`	90.2% → 89.7%	0.047 → 0.046

Table 8: Supervised points per window, two seeds, seed means.

data points per window	shedding retained	relative L_2
0	90.2%	0.047
100	96.2%	0.030
500	97.6%	0.021
2000	99.4%	0.016
4000	100.0%	0.015

Caveat. Ten zero crossings is enough to state St to about a percent, but it is still a shorter record than the LBM's own validation (∼12 periods of a signal that is not itself being predicted). The frequency agreement to +0.1% should be read as "within the resolution of this measurement", not as a claim of three-digit accuracy.

Visible artefact. The probe trace shows small discontinuities at window handoffs, where each window starts from the previous one's end state rather than continuing it smoothly. This is the cost of the mechanism that makes the wake survive, and is reported rather than smoothed away.

Table 9: Prediction accuracy of the best data-free configuration (`all`, $\sigma = 8$) at Re = 100, over one shedding period and over five.

quantity	1 period (4 windows)	5 periods (20 windows)	reference
shedding frequency St	0.1565 (−0.9%)	**0.1627** (+0.1%)	LBM over the same span
wake amplitude	96.5%	94.8%	LBM probe r.m.s.
phase lag	+3.0°	+2.4°	—
phase drift across span	−0.000 periods	−0.005 periods	—
velocity-field error	0.043	**0.036**	relative L_2
zero crossings available	2	10	12+ for the LBM's own St

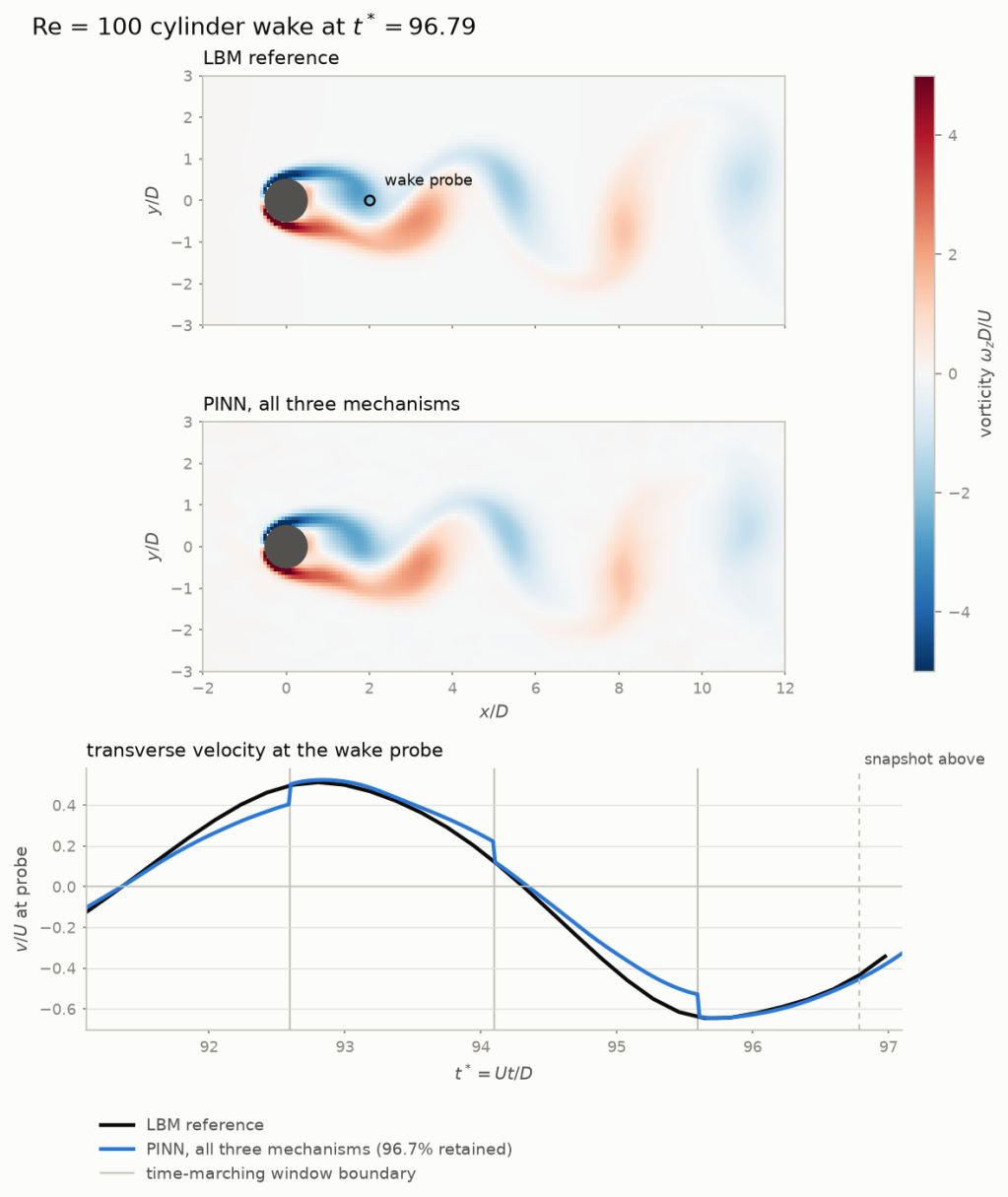


Figure 4: The data-free PINN against the reference. A Kármán street from a network that never saw data. The small steps in the probe trace are the time-marching window handoffs.

7 Does it hold across Reynolds number?

Everything above is at Re = 100. The comparison was therefore repeated at five more Reynolds numbers spanning the two-dimensional shedding range, with fresh ground truth generated and validated at each. The marching window is set to a quarter of each dataset's *measured* shedding period, so all configurations span exactly one period whatever the frequency — otherwise the comparison would not be like-for-like across Re.

Table 10: The ablation repeated across the shedding range. Two seeds per entry; Re = 100 is the main ablation, same configuration.

Re	St vs correlation	`baseline`	`all` (data-free)	relative L_2
60	+0.6%	37.8, 40.1%	**88.5, 91.2%**	0.031
80	−1.1%	22.2, 21.4%	**95.8, 98.5%**	0.022
100	−3.0%	9.8, 11.3%	**96.7, 98.4%**	0.042
120	+2.5%	20.2, 26.5%	**103.5, 104.1%**	0.034
150	−0.3%	35.7, 43.1%	**94.1, 96.7%**	0.032
180	−1.6%	31.1, 38.3%	**93.6, 96.3%**	0.026

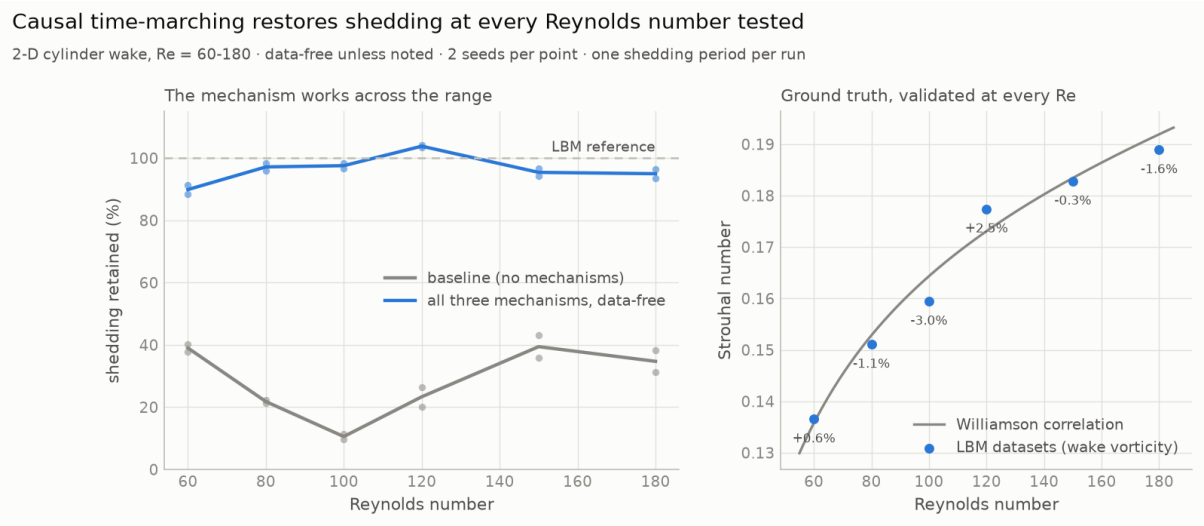


Figure 5: Left: shedding retained against Reynolds number, every seed plotted. Right: each dataset's own validation, measured Strouhal number against Williamson's correlation.

The result generalises. The control keeps 10–43% of the wake and the mechanism-equipped configuration 88–104%, at every Reynolds number tested, with field errors of 2–4%. The claim is therefore about the shedding regime rather than about one dataset.

Three things not to over-read. Re = 120 exceeds 100%: the network's wake is slightly *stronger* than the reference, which is amplitude overshoot, not extra accuracy. The `baseline` column varies from 10% to 43% with no clean trend and a seed spread of 5–8 points, so its variation should not be given a physical reading. And the Re = 60 and 80 datasets run at a lower Mach number than the rest, for the reason in Section 7.1 — closer to incompressible, but not an identical setup.

7.1 An acoustic resonance that captured the low-Re wakes

The first attempt at this sweep produced Strouhal numbers that fell as Reynolds number rose — 0.114 at Re = 60, 0.113 at Re = 80, 0.159 at Re = 100 — which is impossible for this flow, and worth recording as a methodological result.

The cause is not the measurement. Wake *vorticity*, which cannot see irrotational sound, oscillates at the same anomalous frequency. Lattice Boltzmann is weakly compressible and the free-slip walls reflect perfectly, so the domain supports a standing half-wave across its height at $c_s/2H$, with $c_s = 1/\sqrt{3}$ on the D2Q9 lattice. For the sweep geometry ($H = 1200$, $D = 24$, $u = 0.05$) that is $\mathrm{St}_a = 0.1155$ in flow units, and the distance from the natural shedding frequency decides whether the wake locks onto it. Separation is measured as $(\mathrm{St} - \mathrm{St}_a)/\mathrm{St}$, with St taken from Williamson's correlation [12] rather than from the measurement — at the Reynolds numbers that locked, the measured value *is* the locked one, so using it would define the anomaly away:

Re	separation from the resonance	$\mathrm{St}/\mathrm{St}_a$	outcome
60	15%	1.18	locked, wake at 0.114
80	24%	1.32	locked, wake at 0.113
100	30%	1.42	clean, −3.0% against the correlation
120–180	33–40%	1.50–1.66	clean, −1.6% to +2.5%

This is a known physical mechanism, not a numerical curiosity. Free-slip lateral walls are perfectly reflecting duct walls, and lattice Boltzmann carries genuine sound, so the simulation is a duct that resonates. Blevins [15] showed experimentally that a transverse acoustic wave shifts cylinder shedding to its own frequency, and that the entrainment is driven by the acoustic *velocity* rather than the pressure; Hong et al. [14] model the bidirectional vortex–sound coupling for a cylinder in a flow duct and place the lock-in band at $\mathrm{St}/\mathrm{St}_a \leq 1.14$. The ratios above are 1.18 and 1.32 for the two cases that locked — outside that band, and the discrepancy is worth naming rather than hiding. The most likely explanation is that these walls are lossless where a physical duct is not, so the resonance is sustained at separations a real duct could not hold; confirming that would need a run with an absorbing lateral boundary, which is listed in Section 10 as outstanding.

The resonance sits at $(c_s/2H)(D/U)$ in flow units, so *lowering* the lattice velocity moves it away from the shedding band rather than merely weakening it. Regenerating Re = 60 and 80 at $u = 0.0277$ and 0.0280 (against 0.05 elsewhere) moved them to +0.6% and −1.1% against the correlation. The same resonance explains a long-standing oddity in this project: the lift coefficient's r.m.s. never converged between runs, because its amplitude is modulated by the acoustic mode beating against the shedding at *every* Reynolds number, including Re = 100.

Consequently all Strouhal numbers reported here are measured from wake vorticity averaged over many points, with a single-point probe velocity as an independent check — the two agree to about 1% at every Re. The lift signal is retained for C_d, which it measures well, but is no longer trusted for frequency: at Re = 80 it reads 0.1619 where the vorticity says 0.1511 and the correlation 0.1528.

8 Physics in a neural operator

The reference paper's FNO is purely data-driven. This section reproduces its recipe and then asks whether physics improves it.

Reproduction. `underPINN`'s `FNO2D`, trained after Renn et al.: Reynolds-decompose first, map two consecutive frames to the next, apply recursively for ten steps during training with the loss

summed over every step; width 80, 24 modes, Adam at 1×10^{-3}, 200 epochs, batch 16.

Contribution. A physics residual is added to that loss. The network predicts (u, v) only, so any residual containing ∇p is unusable; taking the curl of the momentum equations removes the pressure and leaves vorticity transport,

$$\omega_t + u\,\omega_x + v\,\omega_y - \tfrac{1}{\mathrm{Re}}\nabla^2\omega = 0, \qquad \omega = v_x - u_y, \tag{4}$$

evaluated by finite differences on the grid. Continuity is available separately as the cheaper variant. Both residuals are normalised by the residual the LBM data *itself* leaves — the data is weakly compressible and second-order accurate, so driving a prediction below that floor would be fitting discretisation error, not physics.

Table 11: Rollout error (relative L_2 on (u, v), held-out initial conditions, two seeds). The trained horizon is ten steps.

forecast steps in shedding periods	10 (trained) 0.31	20 0.61	30 0.92	50 1.53
`data` (paper recipe)	0.093	0.183	**0.296**	**0.365**
`physics` (continuity + vorticity)	**0.074**	**0.132**	0.307	0.772
`physics_div` (continuity only)	0.075	0.140	0.312	0.764

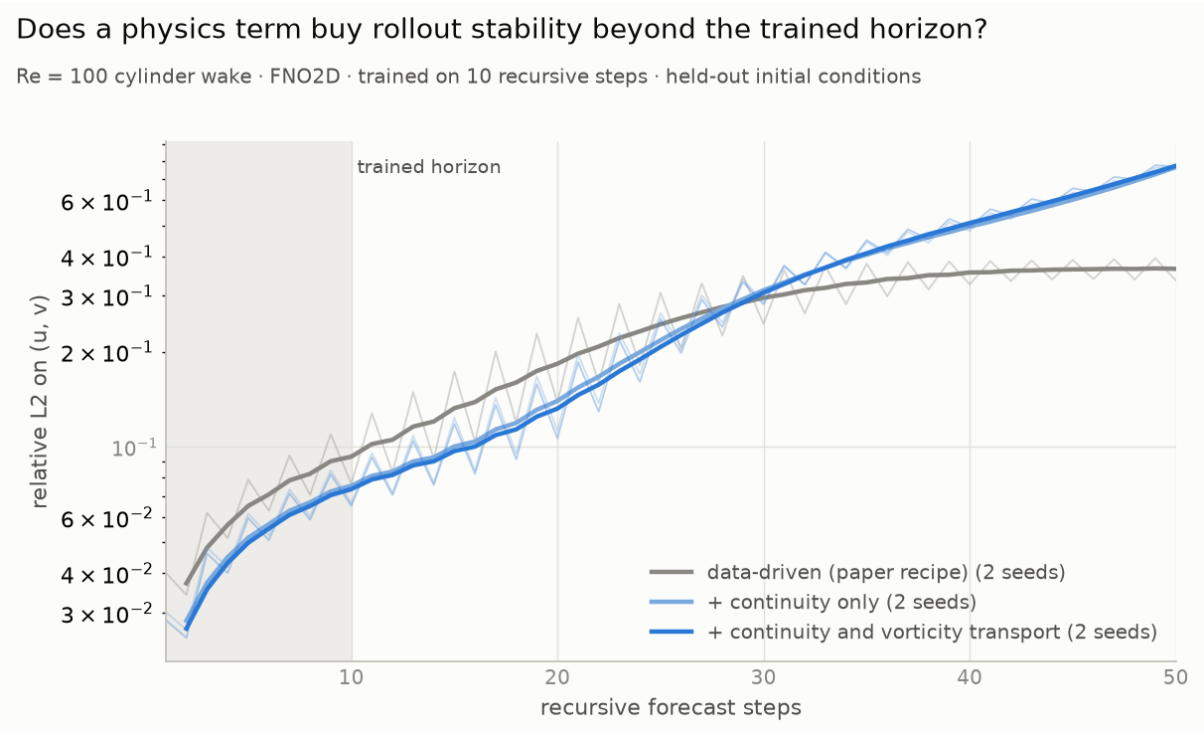


Figure 6: Forecast error against horizon. Bold lines are pairwise means over adjacent steps; faint lines are raw (see the parity note). The curves cross at approximately one shedding period.

Result. Physics lowers forecast error by 21% at the trained horizon and 28% at twice it, then *crosses over* at step 27–32 across seeds and variants, which is 0.8–1.0 shedding periods. Continuity alone buys nearly all of the benefit, so the effect is not specific to the vorticity term — useful, since continuity is the cheaper residual by far.

Why the late-horizon ordering is not reported as a result. By 50 steps both variants have failed in the sense that matters: the error on the fluctuation exceeds unity, meaning that predicting zero fluctuation would beat either rollout. The data-driven curve flattens because it decays towards the mean flow — the safest wrong answer — while the physics-constrained rollouts keep injecting fluctuation energy and overshoot. Across seeds the physics variant reads 0.63 and 0.92 at 50 steps, so the gap is not reproducible. The defensible statement is that *physics extends the useful forecast horizon, and neither model is usable beyond one shedding period.*

A parity artefact worth naming. Rollout error alternates between even and odd steps by 24–44% in *every* variant, physics or not: a period-2 mode of the two-frame autoregressive map. A single step's error can therefore read 30% better or worse purely by which parity it lands on, so all numbers above are pairwise means of adjacent steps, and the figure keeps the raw curve visible.

Comparison with the reference paper. At the trained horizon the data-driven variant reaches $\epsilon = 0.093$ against the paper's $\epsilon < 0.1$. *This is not an improvement on their work.* Our data is clean, periodic, two-dimensional LBM output at $\mathrm{Re} = 100$; theirs is noisy experimental PIV at $\mathrm{Re} = 240$–3060 with genuine three-dimensionality. The only claim supported is that the reproduction works.

9 Conclusions

1. The documented collapse reproduces on validated ground truth: a data-free PINN retains 10.6% of the wake, and quadrupling the training budget does not fix it.
2. **Causal time-marching alone restores shedding, data-free** (89.2% retained), and an equal-budget control attributes the gain to causality rather than to compute. Fourier features and the Reynolds decomposition do not escape the collapse on their own, but improve accuracy substantially once marching is in place, taking the field error from 0.078 to 0.042.
3. The best data-free configuration predicts the shedding frequency to 0.9%, the amplitude to 96.5% and the phase to 3°, over one shedding period, with a velocity-field error of 4.3%.
4. Supervision is needed in far smaller quantity than expected: 100 points per window restores 96% of the shedding, and further data buys accuracy only.
5. Adding a pressure-free physics residual to a data-driven FNO lowers forecast error by 21–28% within one shedding period, with continuity alone responsible for nearly all of it. Beyond one period both variants fail.

10 Limitations and ongoing work

- **Confinement bias in C_d.** The production datasets run at $H = 50$, where Qu et al. [11] report blockage effects are still significant ($H > 100$ is needed). C_d is correspondingly high by 2.7–7.9% at every Reynolds number in the sweep, while St, which the literature reports is far less sensitive to domain size, is within 3.3% throughout. Nothing in this report depends on C_d — every PINN and FNO result is scored against velocity fields and wake probes — but a single run at $H > 100$ would close the validation question outright, and is the most valuable outstanding experiment.

- **The shedding range, not the documented case.** The sweep covers Re = 60–180, its upper limit set by the three-dimensional transition at $\mathrm{Re}_c = 188.5 \pm 1.0$ [13]. The collapse this project responds to was documented at Re = 200 [1], above that transition, so what is reproduced here is the phenomenon rather than their specific case.

- **Two seeds throughout.** Every table is a mean over two seeds, and the spread is not always small: `decompose` gives 9.3% and 4.6% retained shedding, a factor of two, reported as 7.0%. The conclusions rest on effects far larger than that spread, but three to five seeds with a reported spread would remove the objection.

- **Soft window continuity.** The marching chain enforces continuity between windows only through the initial-condition loss, which is what produces the visible handoff steps in Figure 4. Roy and Castonguay [10] enforce it exactly through a solution ansatz, on one-dimensional problems; applying that here is the clearest available extension.

- **Five shedding periods of PINN evaluation.** Section 6 covers five periods, over which the accuracy improves rather than decays. Behaviour over tens of periods, where a marching chain might accumulate phase error, is untested — and the cost grows linearly with the horizon, one window per quarter period.

- C_l **r.m.s. is not converged** across configurations (0.18–0.56 against a benchmark of 0.222–0.235 [11]) and is deliberately not quoted as a validation metric.

- **The resolution series was run at 5% blockage**, where blockage masks resolution. The blockage series is sound; a clean resolution answer would require repeating it at $H = 40D$.

- **An absorbing lateral boundary** would confirm the acoustic mechanism of Section 7.1 rather than sidestepping it. Lowering the lattice velocity moves the resonance out of the shedding band; damping it would remove it.

A note on method. Estimates for each phase were recorded *before* running it. Three were wrong in ways that changed conclusions: the collapse was expected to need all three mechanisms (it needs one); the Fourier bandwidth was expected to be the quick win (it does nothing alone); and the FNO physics term was expected to help only beyond the trained horizon (it helps within it and hurts beyond). An earlier CPU sweep at a fifth of the budget supported the first of those wrong conclusions — *an under-trained ablation produces the wrong qualitative answer, not merely a noisier one.*

References

[1] P.-Y. Chuang and L. A. Barba, *Predictive Limitations of Physics-Informed Neural Networks in Vortex Shedding*, arXiv:2306.00230, 2023.

[2] P. J. Renn, C. Wang, S. Lale, Z. Li, A. Anandkumar and M. Gharib, *Forecasting subcritical cylinder wakes with Fourier Neural Operators*, arXiv:2301.08290, 2023.

[3] D. Shah, *Complementary, Not Cumulative: Interaction Effects in Physics-Informed Neural Networks for Navier–Stokes Vortex Shedding*, arXiv:2608.19632, 2026.

[4] A. S. Krishnapriyan, A. Gholami, S. Zhe, R. M. Kirby and M. W. Mahoney, *Characterizing possible failure modes in physics-informed neural networks*, Advances in Neural Information Processing Systems (NeurIPS), 2021.

[5] S. Wang, S. Sankaran and P. Perdikaris, *Respecting causality is all you need for training physics-informed neural networks*, Computer Methods in Applied Mechanics and Engineering, 2024. doi:10.1016/j.cma.2024.116813.

[6] S. Wang, H. Wang and P. Perdikaris, *On the eigenvector bias of Fourier feature networks: From regression to solving multi-scale PDEs with physics-informed neural networks*, Computer Methods in Applied Mechanics and Engineering **384**, 113938, 2021.

[7] M. Tancik et al., *Fourier features let networks learn high frequency functions in low dimensional domains*, NeurIPS, 2020.

[8] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart and A. Anandkumar, *Fourier Neural Operator for Parametric Partial Differential Equations*, ICLR, 2021.

[9] Z. Li et al., *Physics-Informed Neural Operator for Learning Partial Differential Equations*, ACM/IMS Journal of Data Science, 2024. doi:10.1145/3648506.

[10] P. Roy and S. Castonguay, *Exact Enforcement of Temporal Continuity in Sequential Physics-Informed Neural Networks*, arXiv:2403.03223, 2024.

[11] L. Qu, C. Norberg, L. Davidson, S.-H. Peng and F. Wang, *Quantitative numerical analysis of flow past a circular cylinder at Reynolds number between 50 and 200*, Journal of Fluids and Structures **39**, 347–370, 2013.

[12] C. H. K. Williamson, *Defining a universal and continuous Strouhal–Reynolds number relationship for the laminar vortex shedding of a circular cylinder*, Physics of Fluids **31**(10), 2742, 1988.

[13] D. Barkley and R. D. Henderson, *Three-dimensional Floquet stability analysis of the wake of a circular cylinder*, Journal of Fluid Mechanics **322**, 215–241, 1996.

[14] Z. Hong, X. Wang, X. Jing and X. Sun, *Frequency lock-in mechanism in flow-induced acoustic resonance of a cylinder in a flow duct*, Journal of Fluid Mechanics **884**, A42, 2020.

[15] R. D. Blevins, *The effect of sound on vortex shedding from cylinders*, Journal of Fluid Mechanics **161**, 217–237, 1985.

[16] T. G. Grossmann, U. J. Komorowska, J. Latz and C.-B. Schönlieb, *Can physics-informed neural networks beat the finite element method?*, IMA Journal of Applied Mathematics **89**(1), 143–174, 2024.

[17] Aeroscience Computations and Analysis Lab, `underPINN`: a JAX framework for physics-informed neural networks.